

# Chinatip Lawansuk (廖有福)

Taipei, Taiwan | Email: l.chinatip@gmail.com | Mobile: +886987422475 | LinkedIn: Chinatip Lawansuk

Github: github.com/chinatip-l

## Professional Summary

Driven by a passion for building robust systems from low level, I began my career as an Embedded System Engineer, where I developed firmware and infotainment systems for a fleet of over 1000 electric vehicles. To expand my knowledge from the system level to the chip level, I pursued a Master's degree in Electrical Engineering, focusing on digital IC design. This advanced study allowed me to dive deep into hardware design and optimisation, culminating in a thesis where I designed and implemented a CORDIC hardware accelerator with a custom caching system.

## Expertises

**Programming:** Verilog, SystemVerilog, C, C++, Python, Assembly, Shell Script, TCL, Tensorflow

**IC Design Software:** Synopsys VCS, Synopsys Design Compiler (Design Vision), Synopsys TetraMAX, Cadence NC-Verilog, Cadence Innovus, Siemens Calibre

**Skills:** Cell-Based Digital IC Design Flow, Image Processing, Computer Architecture, Operating System, Embedded System Design, Firmware, RTOS, Circuit Design

**Coursework:** Computer-Aided VLSI System Design, Mixed-Signal IC Design, Computer Vision, Computer Architecture, Computer Organisation, Operating System

## Research and Publication

### CORDIC hardware Improvement using Content Addressable Memory Cache (Thesis)

- Investigated and designed an algorithm to cache partially calculated data and reuse it in the CORDIC architecture for sine and cosine calculation.
- Designed and implemented the processing core to suit the caching algorithm
- Designed and implemented a custom TCAM to use with the proposed cache unit and a complete cache subsystem from scratch, including multiple cache blocks, a PLRU replacement policy, and a controller
- Designed and implemented a pipeline and synchronisation mechanism using a handshaking protocol
- Completed the full cell-based IC design flow from logic design and verification, synthesis, Design For Testability, Automatic Place and Route, post-layout simulation, and transistor-level simulation
- Using TSMC 90nm technology, resulting in a peak throughput of 67.796 MOPS, consumes 48.2 mW @ 338.98 MHz.

### Two-Dimensional Convolution Accelerator of High-Precision Quantization (Conference Paper)

- Y. -T. Chao, J. -Y. Gao, **C. Lawansuk** and Y. -C. Fan, "Two-Dimensional Convolution Accelerator of High-Precision Quantization," *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA, 2024, pp. 1-2, doi: 10.1109/ICCE59016.2024.10444315.
- Developed a high-precision quantized 2D convolution accelerator leveraging the Winograd algorithm, achieving <0.2% quantization error.

## Education

**National Taipei University of Technology (國立臺北科技大學)**

Sept 2023 – Aug 2025

MSc in Electrical Engineering (GPA: 4.00/4.00)

• **Laboratory:** Integrated System Laboratory (ISLAB) @ NTUT

**King Mongkut's Institute of Technology Ladkrabang (KMITL)**

Aug 2017 – Jul 2021

BE in Computer Engineering (GPA: 3.29/4.00)

## Awards

**First-Ranked Graduating Student** (National Taipei University of Technology)

May 2025

**Second Class Honour** (KMITL)

Jul 2021

## Experience

**Firmware Engineer Intern**, LinkCom Manufacturing Co., Ltd. – New Taipei, Taiwan

Aug 2024 - May 2025

- Worked on a high-power-density DC-DC Converter including hardware design (Planar Transformer, LLC topology) and embedded software optimisation. Performance reaches more than 1 kW on the credit card size module.
- Initiate the project for the transformer test platform (Firmware, and Hardware)
- Investigate and design closed loop control system for power module and identify impact parameters

**Embedded System Engineer**, MuvMi Co., Ltd. – Bangkok, Thailand

Oct 2021 - Sept 2023

- Develop IoT Fleet, Data Analysis and Visualisation Platform on AWS for EV Charging System, identified and mitigated up to 12% of energy loss
- Develop a Vehicle Control Unit Firmware (RTOS on arm Cortex M4 processor) and infotainment system (Buildroot on arm Cortex A7 processor) for Electric Minibus and Electric Tuk-Tuk for the fleet of 1000 vehicles

# 國立臺北科技大學 碩士學位證書

(114)北科大碩證字第1583號

學生 廖有福

中華民國捌拾捌年貳月拾陸日生

在本校電資學院外國學生專班碩士班研究期滿經碩士  
學位考試合格依學位授予法之規定

授予工學碩士學位

此證

院長 張陽郎

校長 王錫福



中華民國

壹佰壹拾肆年

陸月

日



外籍生國籍：泰國  
學號：112998405





國立臺北科技大學

# National Taipei University of Technology

*The President of National Taipei University of Technology,  
on the Recommendation of the Graduate Council, Has Conferred upon*

**Chinatip Lawansuk**

*Who Has Satisfactorily Fulfilled All Requirements for  
the Degree of*

## MASTER OF SCIENCE

*in Electrical Engineering and Computer Science,  
with All the Rights, Privileges, and Honors Thereunto Appertaining,  
in Witness Whereof the Seal of the University and the Signature of*

*the Proper Authority Is Hereunto Affixed.*

*Given in Taipei, Taiwan, Republic of China*

*The Thirtieth of June, in the Year of Two Thousand and Twenty Five*

*Bo-fu Wang*

PRESIDENT OF THE UNIVERSITY

*Yeh-Shyan Kwang*

PROVOST OF ACADEMIC AFFAIRS

Student ID NO. 112998405



NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY  
Taipei, Taiwan, R. O. C.

Name: Chinatip Lawansuk(Chinatip Lawa)

Date of Birth: Feb. 16, 1999

Date Enrolled: Sep. 2023

Date Conferred: \*\*\*\* \*

Date Issued: Feb. 20, 2025

Graduate Institute: International Program of Electrical Engineering and Computer Science(IEECS)

Degree Conferred: \*\*\*\* \*

The following transcript is hereby certified as correct according to the record of the university.

Student ID: 112998405

| Course                                                             | 1st Semes. |       | 2nd Semes. |       | Course | 1st Semes. |       | 2nd Semes. |       | Course | 1st Semes. |       | 2nd Semes. |       |
|--------------------------------------------------------------------|------------|-------|------------|-------|--------|------------|-------|------------|-------|--------|------------|-------|------------|-------|
|                                                                    | Credit     | Grade | Credit     | Grade |        | Credit     | Grade | Credit     | Grade |        | Credit     | Grade | Credit     | Grade |
| ( 2023 - 2024 )                                                    |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| Graduate Seminars                                                  | 1.0#       | 94    | 1.0#       | 92    |        |            |       |            |       |        |            |       |            |       |
| Detection and Estimation                                           | 3.0#       | 91    |            |       |        |            |       |            |       |        |            |       |            |       |
| Computer-Aided VLSI System Design and Practice                     | 3.0#       | 92    |            |       |        |            |       |            |       |        |            |       |            |       |
| Operating Systems                                                  | 3.0#       | 100   |            |       |        |            |       |            |       |        |            |       |            |       |
| Computer Vision                                                    | 3.0#       | 90    |            |       |        |            |       |            |       |        |            |       |            |       |
| Deep Learning for Digital Image Analysis                           |            |       | 3.0#       | 88    |        |            |       |            |       |        |            |       |            |       |
| Error Control Technologies for Modern Mobile Communication Systems |            |       | 3.0#       | 96    |        |            |       |            |       |        |            |       |            |       |
| Mixed-Signal Integrated Circuit Design                             |            |       | 3.0#       | 91    |        |            |       |            |       |        |            |       |            |       |
| Data Science Principles with Applications on Educational Data      |            |       | 3.0#       | 87    |        |            |       |            |       |        |            |       |            |       |
| Conduct                                                            | ---        | 86    | ---        | 85    |        |            |       |            |       |        |            |       |            |       |
| Total Credits & Average ( 2024 - 2025 )                            | 13.0       | 93.3  | 13.0       | 90.6  |        |            |       |            |       |        |            |       |            |       |
| Master's Thesis                                                    | 3.0#       |       | 3.0#       |       |        |            |       |            |       |        |            |       |            |       |
| Conduct                                                            | ---        | 85    | ---        |       |        |            |       |            |       |        |            |       |            |       |
| Total Credits & Average                                            | 0.0        | ---   |            |       |        |            |       |            |       |        |            |       |            |       |
| Thesis                                                             |            |       |            | ****  |        |            |       |            |       |        |            |       |            |       |
| Total Credits "26.0"                                               |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| Total Average "91.96"                                              |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| GPA "4.00"                                                         |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| Rank in Department : 1/8                                           |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| Total EMI Credits : 26.0                                           |            |       |            |       |        |            |       |            |       |        |            |       |            |       |
| TO BE CONTINUED                                                    |            |       |            |       |        |            |       |            |       |        |            |       |            |       |

Remarks: The grading system is as follows: 80 or more=A; 70 to 79=B; 60 to 69=C; 50 to 59=D; less than 50=E; 70=the passing grade, W=Withdrawal, CW=Course Waived, P=Passing, F=Failing, #=English as a Medium of Instruction.



Signature:

Wan-Ting Huang

Registrar

Signature:

Yueh-Shyan Huang

Provost of Academic Affairs

B157238





# KING MONGKUT'S INSTITUTE OF TECHNOLOGY LADKRABANG

Chalongkrung Road, Ladkrabang, Bangkok 10520, THAILAND

## TRANSCRIPT OF RECORDS

Faculty of Engineering

Name : Mr. Chinatip LAWANSUK

Date of Birth : February 16, 1999

Program : Computer Engineering

Date of Admission : August 7, 2017

Student ID : 60010235

Degree : Bachelor of Engineering (Second Class Honors)

Major : -

Date of Graduation : June 14, 2021

| Course                                                        | Credit | Grade | Course                                                 | Credit | Grade |
|---------------------------------------------------------------|--------|-------|--------------------------------------------------------|--------|-------|
| <b>1st Semester, Academic Year 2017</b>                       |        |       | <b>Special Semester, Academic Year 2019</b>            |        |       |
| 01006028 PRE-ACTIVITIES FOR ENGINEERS                         | 1      | A     | 01006004 INDUSTRIAL TRAINING                           | 0      | S     |
| 01006030 CALCULUS 1                                           | 3      | A     | GPS : - GPA : 3.20                                     |        |       |
| 01076001 INTRODUCTION TO COMPUTER ENGINEERING                 | 3      | B+    | <b>1st Semester, Academic Year 2020</b>                |        |       |
| 01076002 PROGRAMMING FUNDAMENTAL                              | 3      | B+    | 01076015 COMPUTER ENGINEERING PROFESSIONAL DEVELOPMENT | 2      | A     |
| 90201001 FOUNDATION ENGLISH 1                                 | 3      | A     | 01076311 PROJECT 1                                     | 3      | B+    |
| 90306003 LIVING SKILLS                                        | 3      | A     | 90401013 GENERAL BUSINESS                              | 3      | A     |
| GPS : 3.81 GPA : 3.81                                         |        |       | 90591008 PHOTOGRAPHY APPRECIATION                      | 3      | C+    |
| <b>2nd Semester, Academic Year 2017</b>                       |        |       | GPS : 3.45 GPA : 3.22                                  |        |       |
| 01006031 CALCULUS 2                                           | 3      | B     | <b>2nd Semester, Academic Year 2020</b>                |        |       |
| 01076003 CIRCUITS AND ELECTRONICS                             | 4      | C+    | 01076024 SOFTWARE ARCHITECTURE AND DESIGN              | 4      | A     |
| 01076004 OBJECT ORIENTED PROGRAMMING                          | 3      | B+    | 01076312 PROJECT 2                                     | 3      | B+    |
| 01076012 DISCRETE STRUCTURE                                   | 3      | A     | 90101002 MATHEMATICS IN DAILY LIFE                     | 3      | A     |
| 90201002 FOUNDATION ENGLISH 2                                 | 3      | B+    | 90108007 ENVIRONMENTAL STUDY                           | 3      | A     |
| 90201030 ENGLISH FOR PROFESSIONAL PRESENTATION                | 3      | B+    | GPS : 3.88 GPA : 3.29                                  |        |       |
| 90401011 ENTREPRENEURSHIP                                     | 3      | A     | Total Credits Earned : 135                             |        |       |
| GPS : 3.38 GPA : 3.56                                         |        |       | Cumulative GPA : 3.29                                  |        |       |
| <b>1st Semester, Academic Year 2018</b>                       |        |       | ----- End of Transcript -----                          |        |       |
| 01006032 ELEMENTARY DIFFERENTIAL EQUATIONS AND LINEAR ALGEBRA | 3      | B     |                                                        |        |       |
| 01076005 DATA STRUCTURES AND ALGORITHM                        | 3      | C+    |                                                        |        |       |
| 01076006 DIGITAL SYSTEM FUNDAMENTALS                          | 4      | B+    |                                                        |        |       |
| 01076007 DATA COMMUNICATIONS                                  | 4      | C+    |                                                        |        |       |
| 90020008 ENGLISH FOR MARKETING                                | 3      | A     |                                                        |        |       |
| GPS : 3.08 GPA : 3.41                                         |        |       |                                                        |        |       |
| <b>2nd Semester, Academic Year 2018</b>                       |        |       |                                                        |        |       |
| 01076008 SOFTWARE DEVELOPMENT PROCESSES                       | 4      | C+    |                                                        |        |       |
| 01076009 COMPUTER ORGANIZATION AND ASSEMBLY LANGUAGE          | 4      | C+    |                                                        |        |       |
| 01076010 COMPUTER NETWORKS                                    | 4      | B+    |                                                        |        |       |
| 01076253 PROBABILITY AND STATISTICS                           | 3      | B+    |                                                        |        |       |
| GPS : 2.96 GPA : 3.32                                         |        |       |                                                        |        |       |
| <b>1st Semester, Academic Year 2019</b>                       |        |       |                                                        |        |       |
| 01076011 OPERATING SYSTEMS                                    | 3      | B+    |                                                        |        |       |
| 01076021 COMPUTER ARCHITECTURE                                | 4      | C+    |                                                        |        |       |
| 01076022 MICROCONTROLLER APPLICATION AND DEVELOPMENT          | 3      | B+    |                                                        |        |       |
| 01076026 USER EXPERIENCE AND USER INTERFACE DESIGN            | 3      | C+    |                                                        |        |       |
| 01076263 DATABASE SYSTEMS                                     | 3      | C+    |                                                        |        |       |
| 90593015 FUN WITH AI                                          | 3      | B+    |                                                        |        |       |
| GPS : 2.97 GPA : 3.24                                         |        |       |                                                        |        |       |
| <b>2nd Semester, Academic Year 2019</b>                       |        |       |                                                        |        |       |
| 01076013 THEORY OF COMPUTATION                                | 3      | B     |                                                        |        |       |
| 01076014 COMPUTER ENGINEERING PROJECT PREPARATION             | 1      | A     |                                                        |        |       |
| 01076023 COMPUTER HARDWARE DESIGN                             | 3      | B     |                                                        |        |       |
| 01076025 SOFTWARE STUDIO                                      | 3      | B     |                                                        |        |       |
| 01076421 EMBEDDED SYSTEM DESIGN                               | 3      | B+    |                                                        |        |       |
| 01076533 DEEP LEARNING                                        | 3      | B     |                                                        |        |       |
| 01076582 ARTIFICIAL INTELLIGENCE                              | 3      | C+    |                                                        |        |       |
| 90307001 THAI USAGE FOR COMMUNICATION                         | 3      | B     |                                                        |        |       |
| GPS : 3.04 GPA : 3.20                                         |        |       |                                                        |        |       |

Date of Issued : November 7, 2022

Not valid without seal

*Pakorn W.*

(Asst.Prof. Dr. Pakorn Watanachaturaporn)  
Acting Director  
Office of the Registrar





LISTENING AND READING TEST  
OFFICIAL INSTITUTIONAL SCORE REPORT

BA 0118988



Name CHINATIP LAWANSUK

Date of Birth FEBRUARY 16, 1999

ID Number 1100501503550

Test Date NOVEMBER 7, 2022

Client PERSONAL

TOEIC® Services Thailand -- Certified Score Report -- Issued NOVEMBER 8, 2022

LISTENING

460

TOTAL  
SCORE

875

READING

415

Report is valid  
for two years  
from the test  
administration  
date.



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IN1001-01TH



# 國立臺北科技大學獎狀

北科大教字第 1140100181 號

學生 Chinatip Lawa 於 113 學年度  
電資學院外國學生專班應屆  
畢業學業成績 第一名 特頒獎狀  
以資鼓勵

校長 王錫福

中華民國 114 年 6 月 7 日

100 Years of Excellence Cultivating Entrepreneurs of Tomorrow

